

Hariharasudan Ravichandran

Electrical and Electronic Engineering Undergraduate

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PROFILE

Final-year Electrical and Electronic Engineering undergraduate specializing in embedded systems, real-time applications, and FPGA-based digital design. Experienced in ESP32-based firmware development, multi-sensor interfacing, Verilog HDL design using Xilinx Vivado, and PCB design using KiCad. Strong understanding of integrated hardware–firmware co-design and reliable embedded product development.

TECHNICAL SKILLS

Programming: C, Verilog HDL, Python, MATLAB/Octave

Digital Design Platforms: Xilinx Basys3 (Artix-7 FPGA)

Microcontrollers: ESP32, ESP32-S3, ATmega328p

Tools: Xilinx Vivado, KiCad, PSCad, Git, Arduino IDE, Proteus, FreeCAD

PROJECTS

Real-Time Sound Source Localization | [GitHub](#)

Apr 2025 – Present

Developed a system to accurately localize and track sound sources in 2D using a microphone array, TDoA estimation, and camera-based visual cues for enhanced robustness in noisy environments.

ESP32-S3, I2S, DSP, MATLAB, YOLOv8

- Designed a real-time sound localization system using multiple **I2S** microphones
- Implemented **TDoA estimation** using cross-correlation and **GCC-PHAT**
- Applied FIR filtering and exponential smoothing to stabilize angular output
- Implemented frame-based windowing, FFT-based spectral analysis, normalization, and peak detection
- Integrated audio localization output with **YOLOv8**-based visual object detection and camera grid mapping

Smart Bike Safety System | [GitHub](#)

Aug 2025 – Dec 2025

Developed a master-slave embedded system for bike safety: master tracks via GPS and sends crash alerts; slave monitors alcohol and engine, cutting off on detection or unauthorized use.

ESP32, Embedded C, Multi-Node Embedded System, KiCad, FreeCad

- Designed a master-slave ESP32-based bike safety system for accident detection, theft prevention, and emergency alerting
- Implemented crash detection using accelerometer data via **I2C**
- Integrated GPS and GSM modules over **UART** for real-time tracking and SMS alerts
- Implemented relay-based engine cut-off during theft or alcohol detection
- Developed slave-node firmware interfacing **DHT22** and **MQ-3** sensors
- Designed and routed **2-layer PCBs** and performed hardware bring-up
- Created 3D enclosures and mechanical layouts using **FreeCAD**

FPGA-Based Edge Detection | [GitHub](#)

Jan 2026 – Feb 2026

Designed and verified a real-time hardware accelerator for Sobel edge detection targeting the Basys3 FPGA, implementing pixel-wise convolution and gradient magnitude computation in Verilog.

Verilog HDL, Xilinx Vivado, Python, OpenCV, Digital Image Processing

- Implemented a synthesizable 3×3 convolution engine to compute horizontal and vertical image gradients in hardware.
- Developed a bit-accurate Python/OpenCV reference model to generate golden test vectors for functional verification.
- Reduced hardware resource utilization by replacing multipliers and square-root operations with shift-based and absolute-value approximations.
- Validated functionality in Vivado behavioral simulation, achieving 100% match with the software reference output.

Asynchronous FIFO with Robust Clock Domain Crossing | [GitHub](#)

Jan 2026 – Feb 2026

Designed and verified a fully asynchronous FIFO to safely transfer data between independent clock domains, implementing Gray-coded pointers, dual-flop synchronization, and robust full/empty detection logic in Verilog.

Verilog HDL, FPGA Architecture, Digital Design Verification

- Implemented independent read and write clock domains with binary and Gray-coded pointer architectures to ensure safe multi-bit clock domain crossing.
- Designed and integrated dual-flip-flop synchronizers for Gray pointers to mitigate metastability and ensure reliable full and empty flag generation.
- Developed precise full and empty detection logic using next-state Gray pointer comparison with MSB inversion to prevent overflow and underflow conditions.
- Built a self-checking Verilog testbench with asynchronous clocks and scoreboard-based verification to validate data integrity, ordering, and CDC robustness.

Auto Power Factor Correction | [GitHub](#)

Jan 2025 – Apr 2025

Arduino-based APFC system for real-time monitoring and automatic correction of AC load power factor using voltage/current sensors and capacitor bank switching.

Arduino, Voltage & Current Sensors

- Designed an automatic power factor correction system using relay-switched capacitor banks
- Implemented zero-crossing detection for real-time phase difference estimation
- Controlled capacitor banks dynamically under varying load conditions
- Developed and tested embedded firmware in C

DC–DC Boost Converter Design | [GitHub](#)

Dec 2025 – Present

Designing a 12V → 24V DC–DC boost converter with 30W power rating. Completed converter design and open-loop simulation, including component sizing, inductor/capacitor calculations, and waveform validation; voltage-controlled implementation and hardware prototyping ongoing.

Power Electronics, PLECS

- Completed converter design calculations, including duty cycle, inductor, and capacitor sizing.
- Implemented open-loop simulation and verified inductor current and output voltage ripple at multiple load levels.
- Preparing voltage-controlled implementation and hardware prototype for full evaluation.

EDUCATION

B.Sc. Engineering (Hons.) in Electrical and Electronic Engineering

2022 – Present

University of Peradeniya

Sri Lanka

- CGPA: 3.51 / 4.00

G.C.E. Advanced Level (Physical Science Stream)

2020

Hindu College, Colombo

- Results: 2AB | Z-Score - 1.9885

WORK EXPERIENCE

HR Administrative Assistant – Maritime Industry

Jul 2021 – Dec 2021

N-Force Recruitment Services | Full time

VOLUNTEERING EXPERIENCE

Graphic Designing Panelist

Jan 2024 – Present

Perabeats, University of Peradeniya

REFERENCES

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